

RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

CR23532- Cyber and Digital Forensics

LAB MANUAL

2023 REGULATION

FIFTH SEMESTER

III YEAR

COLLEGE VISION

- To be an institution of excellence in Engineering, Technology and Management, Education & Research.
- To provide competent and ethical professionals with a concern for society.

COLLEGE MISSION

- To impart quality technical education imbibed with proficiency and humane values.
- To provide right ambience and opportunities for the students to develop into creative, talented and globally competent professionals.
- To promote research and development in technology and management for the benefit of the society.

DEPARTMENT VISION

To promote highly ethical and innovative cybersecurity professionals through excellence in teaching, training, and research.

DEPARTMENT MISSION

- To produce globally competent cybersecurity experts, motivated to learn emerging technologies and be innovative in solving real-world security challenges.
- To foster research activities among students and faculty that enhance societal security and resilience.
- To impart moral and ethical values in their profession, ensuring a strong commitment to cybersecurity ethics and responsible practices.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of networking, security, data mining, web technology, mobile communication and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry requirements.

PROGRAMME OUTCOMES (POs)

PO1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, naturalsciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leaderin diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leaderin a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage inindependent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Computer Science and Engineering Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze, and develop computer programs in the areas related to algorithms, system software, web design, machine learning, data analytics, networking, and cybersecurity for efficient design of computer-based systems of varying complexity. Familiarity and practical competence with a broad range of programming languages, open-source platforms, and cybersecurity tools and practices.

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational tasks, model real-world problems using appropriate data structures and suitable algorithms, and address cybersecurity challenges. Understanding of standard practices and strategies in software project development, incorporating secure coding practices and using open-ended programming environments to deliver quality and secure products.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and provide solutions to new ideas, with a particular emphasis on cybersecurity challenges. Inculcating a passion for higher studies, creating innovative career paths to be an entrepreneur, and

evolving as an ethically and socially responsible computer science professional with strong cybersecurity expertise.

Subject Code	Subject Name (Lab Oriented Theory course)	Category	L	T	P	C
CR23532	Cyber and Digital Forensics	PC	3	0	2	4

Objectives:	
•	Have an introduction into the process of Cyber Forensics.
•	Understand the Environment of forensics.
•	Learn process of collecting evidences.
•	Gain working knowledge of analyzing evidences using tools.
•	Learn other sources of evidences and its future challenges.

UNIT I	DIGITAL FORENSICS PROCESS	9
Forensic Science, Digital Forensics, Digital Evidence, Digital Forensics Process – Identification, Collection, Examination, Analysis, Presentation Phases. Cyber Crime Law- International Legal Framework of Cybercrime Law, Digital Crime, Investigation Methods for Collecting Digital Evidence.		
UNIT II	**FORENSICS ENVIRONMENT**	**9**
Hardware and Software Environments – Storage Devices, Operating System, File Systems, Metadata, Locating evidence in file systems-Password security, Encryption, and Hidden files. Case study –linking the evidence to the user, Data Analysis using forensics tool ILookIX		
UNIT-III	**COLLECTING EVIDENCES**	**9**
Use of Digital Evidence, File Metadata and Correlation with Other Evidence, Technical Complexities of Digital Evidence. Data carving, Date and time problems, Physical Acquisition and Safekeeping of Digital Evidence. Forensic Imaging Processes. Case Study – IXImager, Understanding ASB container		
UNIT IV	**ANALYZING EVIDENCES**	**9**
Selecting and Analyzing Digital Evidence - Locating digital evidence, categorizing files, eliminating superfluous files, The Event Analysis tool, Cloud Analysis tool, The Lead Analysis tool, Volume Shadow Copy analysis tools, Validating the Evidence. Case study – illustrating the recovery of deleted evidence held in volume shadows.		
UNIT-V	**OTHER SOURCES OF EVIDENCES**	**9**
Windows and Other Operating Systems as Sources of Evidence, Examining Browsers, E-mails, Messaging Systems, and Mobile Phones, Internet and Cloud. -Challenges in Digital Forensics.		
	Total Contact Hours :	**45**

List of Experiments	
1	Study of Computer Forensics and different tools used for forensic investigation
2	Live Forensics Case Investigation using Autopsy
3	Recover Deleted Files using Forensics Tools
4	Find Last Connected USB on your system (USB Forensics)
5	View Last Activity of Your PC
6	Extraction of Browser Artifacts
7	Comparison of two Files for forensics investigation by Compare IT software
8	Collecting Email Evidence in Victim PC
9	Study the steps for hiding and extract any text file behind an image file/ Audio file using Command Prompt.

10	Extract Exchangeable image file format (EXIF) Data from Image Files using Exifreader Software	
	Contact Hours :	**30**
	Total Contact Hours :	**75**

Course Outcomes:	
On completion of course you will be able to	
•	Identify the need for cybercrime investigation.
•	Understand the hardware and software components responsible for seeking evidence.
•	Have knowledge on the techniques used for collecting evidences.
•	Analyse the evidence through suitable tools.
•	Examine other sources of evidences.

Suggested Activities:	
•	Assignment problems.
•	Class presentation/Discussion

Textbooks:	
1.	Richard Boddington, Practical Digital Forensics, First Edition, , PACKT publishing, 2016

Reference Books (s)/Web links:	
1.	John R.Vacca, Computer Forensics, Second Edition, Cengage Learning, 2005.
2.	Richard E.Smith, Internet Cryptography, Third Edition, Pearson Education, 2008.
3.	Marjie T.Britz, Computer Forensics and Cyber Crime: An Introduction, Third Edition, Prentice Hall, 2013.

CO-PO-PSO Matrices of course

PO/PSO CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CR23532.1	3	2	-	-	1	2	1	3	2	1	-	-	2	-	1
CR23532.2	3	2	2	1	3	1	-	-	-	-	-	-	3	2	-
CR23532.3	2	2	2	1	3	1	-	-	-	1	-	-	3	3	-
CR23532.4	1	3	3	3	3	-	-	1	2	2	-	-	2	3	2
CR23532.5	2	3	2	3	3	2	1	1	1	-	-	1	1	3	3
Average	2.2	2.4	2.2	2	2.6	1.5	1	1.6	1.6	1.3	-	1	2.2	2.7	2

EXPERIMENT-01

Aim :

Study of Computer Forensics and different tools used for forensic investigation

What Is Digital Forensics?

Digital forensics is the field of determining who was responsible for a digital intrusion or other computer crime. It uses a wide range of techniques to gain attribution to the perpetrator.

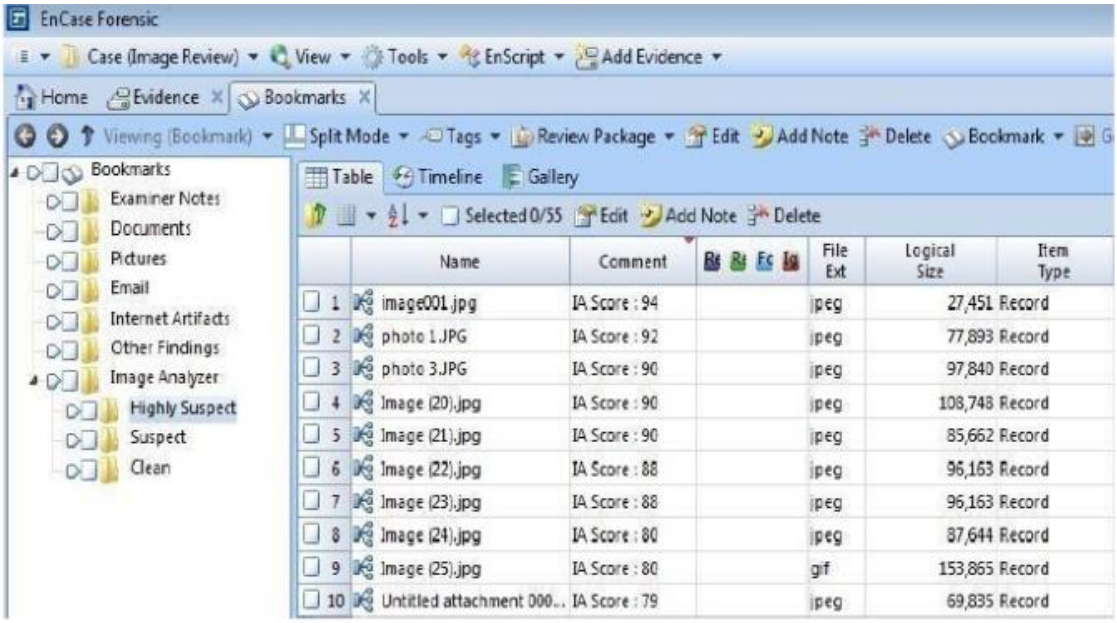
It relies upon the fundamental concept that whenever a digital intrusion or crime is committed, the perpetrator inadvertently leaves a bit of themselves behind for the investigator to find. These "bits" could be entries in log files, changes to the registry, hacking software, malware, remnants of deleted files, etc. All of these can provide clues and evidence to determine their identity and lead to the capture and arrest of the hacker.

As a hacker, the more you know and understand about digital forensics, the better you can evade the standard forensic techniques and even implement anti-forensic measures to throw off the investigator.

The Digital Forensic Tools

Just like in hacking, there are a number of software tools for doing digital forensics. For the hacker, becoming familiar with these tools and how they work is crucial to evading them. Most digital forensic investigators rely upon three major commercial digital forensic suites.

1. Guidance Software's EnCase Forensic
2. Access Data's Forensic Tool Kit (FTK)
3. Prodiscover



These three suites are comprised of multiple tools and reporting features and can be fairly expensive. While these suites are widely used by law enforcement, they use the same or similar techniques as the free open-source suites without the fancy interfaces. By using the open-source and free suites, we can come to understand how such tools as EnCase work without the expense. EnCase is the most widely used tool by law enforcement, but not necessarily the most effective and sophisticated. These tools are designed for user-friendliness, efficiency, certification, good training, and reporting. There are a number of the free, open-source forensic suites, including the following three.

1. The Sleuthkit Kit (TSK)
2. Helix
3. Knoppix



The Forensic Tools Available in BackTrack

In addition, there are a large number of individual tools that are available for digital forensics, some of which are available in our BackTrack and Kali distributions



Some of the better tools in BackTrack include the following, among many others

- sleuthkit
- truecrypt
- hexedit
- autopsy
- iphoneanalyzer
- rifiuti2
- ptk
- exiftool
- evtparse.pl
- fatback
- scalpel
- dc3dd
- driftnet
- timestomp

Forensics Field Tools

- **FTKImager**
  Forensic disk imager and file recovery.

- **Log Parser Lizard GUI**
  Flexible and powerful log file parser. It also does much much more.

- **Noxcivis Field Toolkit**
  The Noxcivis Field Toolkit (NFT) is a free and open interface that allows forensic examiners and collection teams to collect information from a computer.

- **Active@ Partition Recovery**
  Recover deleted partitions.

- **Autopsy :**

  Forensics tool. Autopsy is a digital forensics platform and graphical interface to The Sleuth Kit® and other digital forensics tools. It can be used by law enforcement, military, and corporate examiners to investigate what happened on a computer. You can even use it to recover photos from your camera's memory card.

Result:
Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

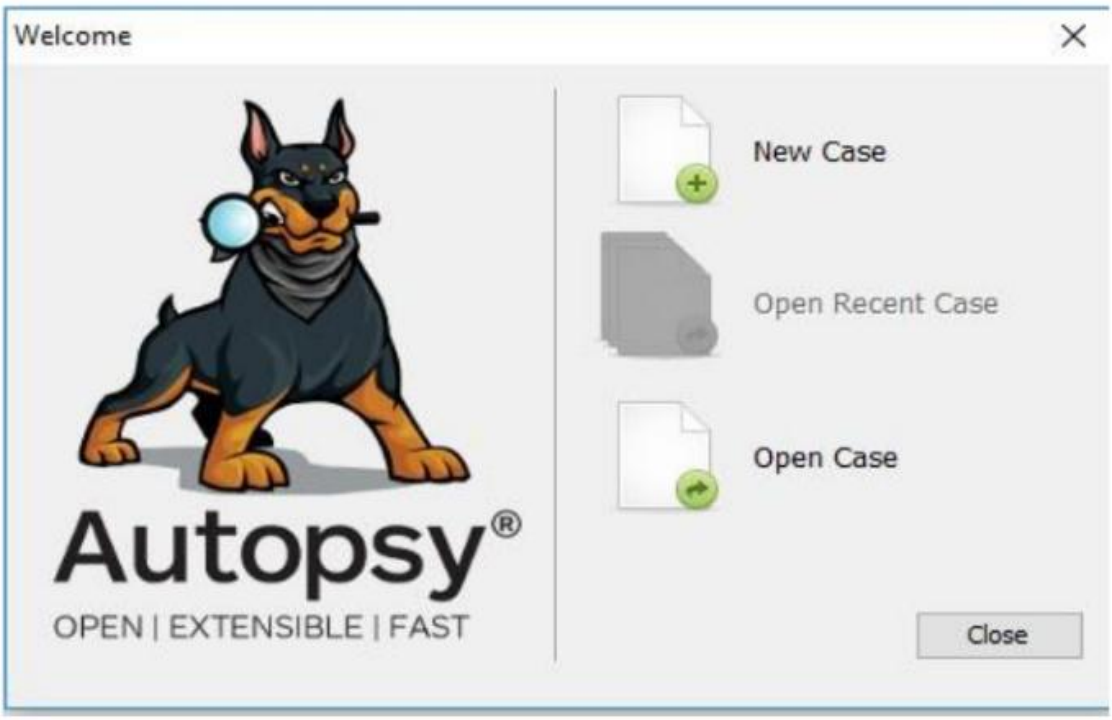
EXPERIMENT-2

Aim :

Live Forensics Case Investigation using Autopsy

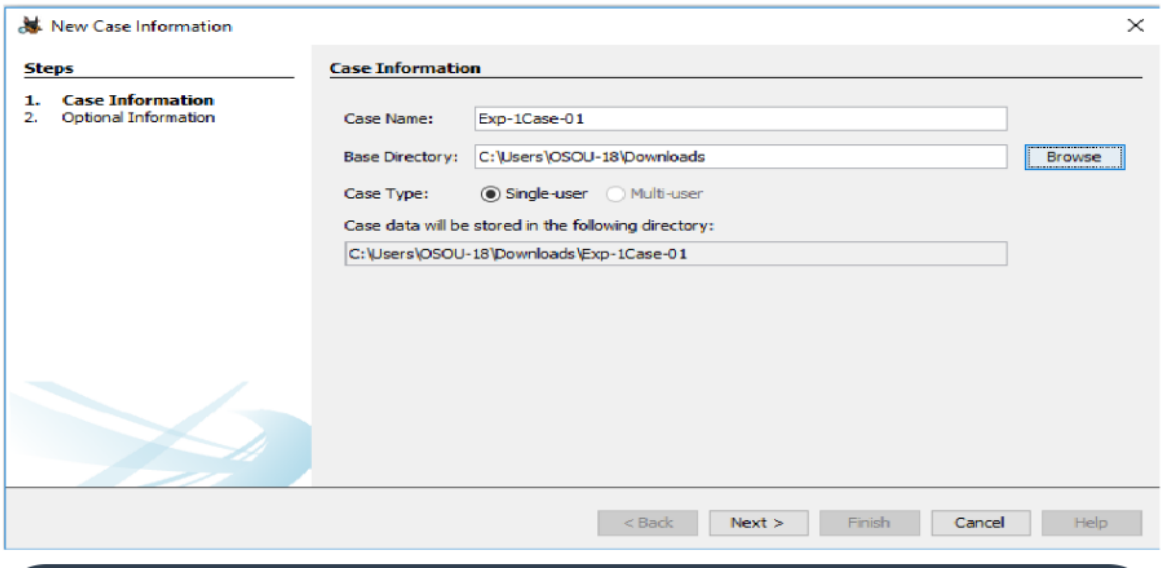
Procedure :

1. First Download autopsy from here and install in your pc. Click ‘New Case’ option.

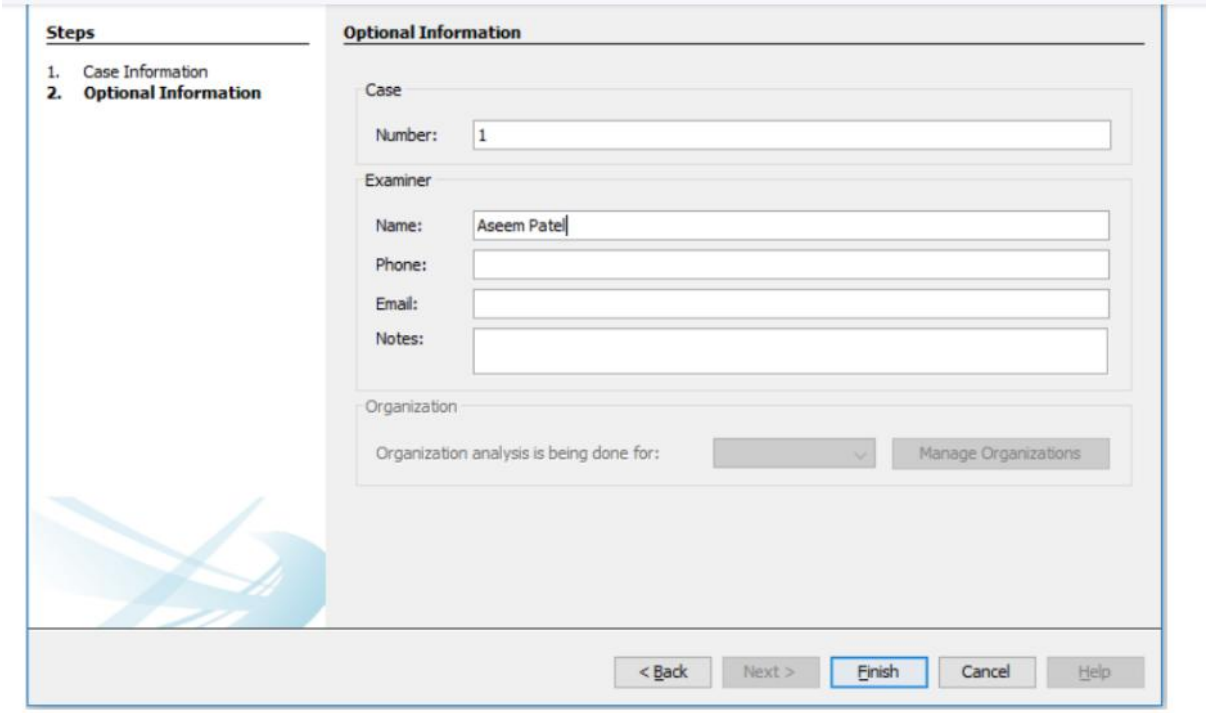


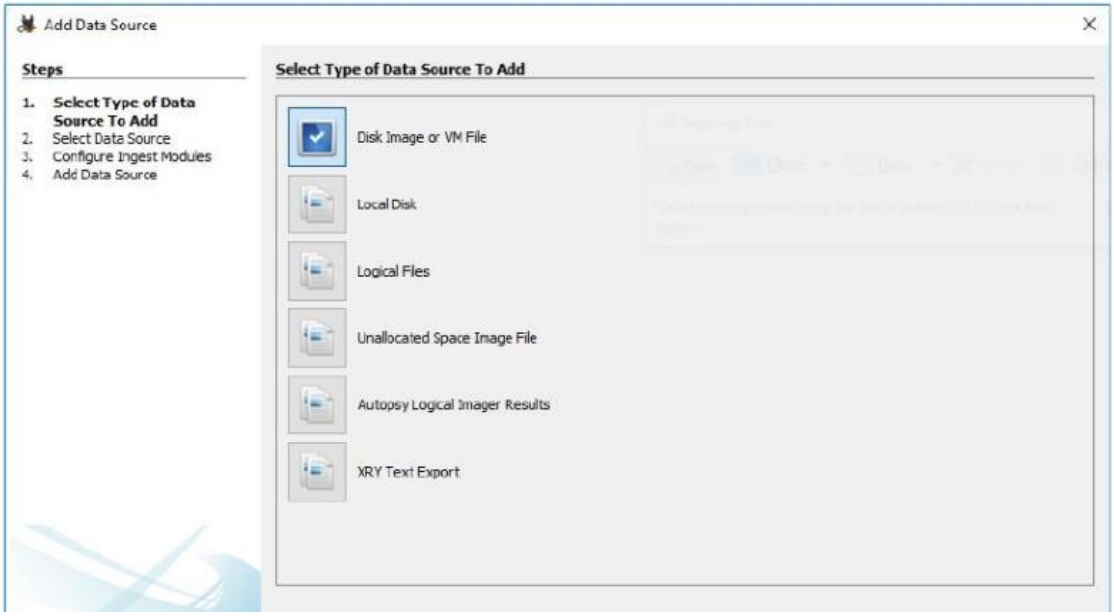
2. A new page will open. Enter the details in ‘Case Name’ and ‘Base Directory’ and choose the location to save the report e.g. :Autoreport.

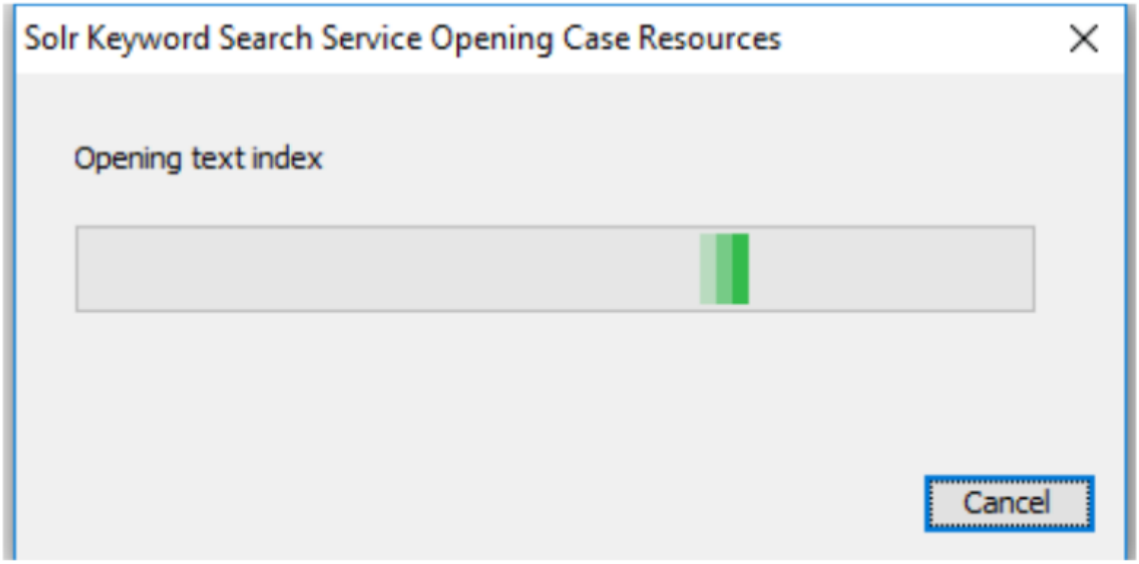
3. Then click on next to proceed to the next step.



Here in the next step, you have to enter the case number and Examiner details and click on finish to proceed to the next step.

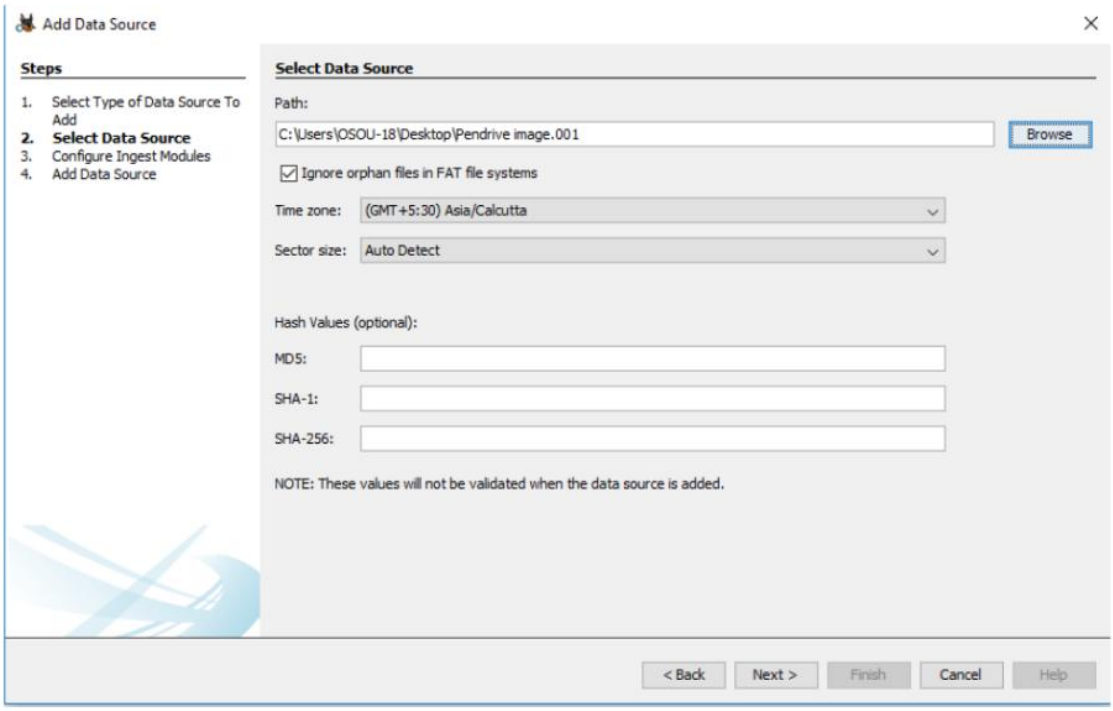




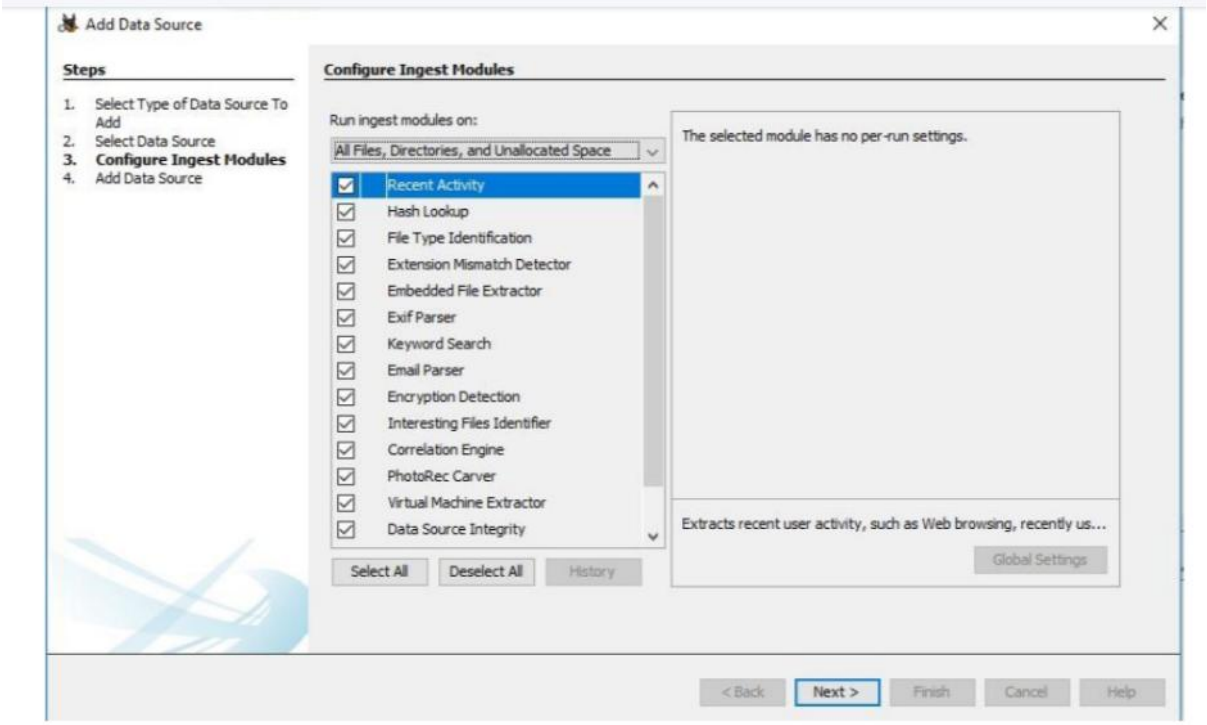


A new window will open. It will ask for the add data source in Step 1. Select source type to add

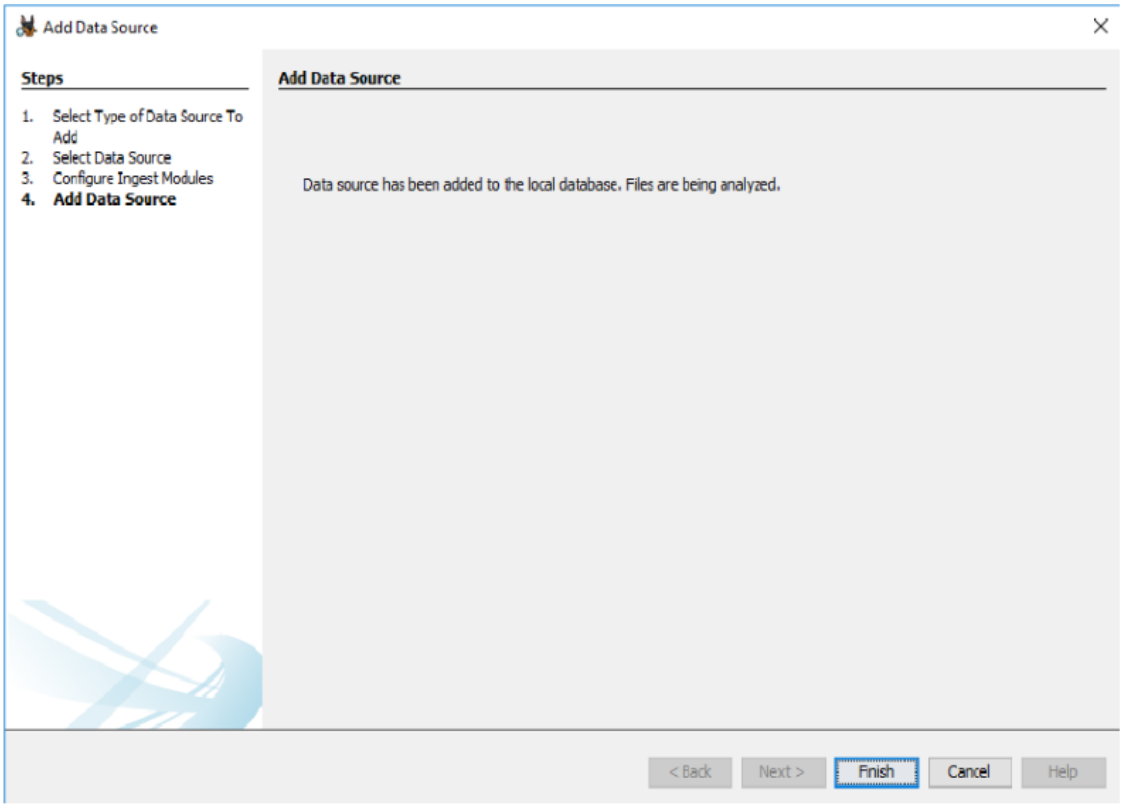
& browse the file Path and click on NEXT option to proceed further.



Configure ingest Modules I have chosen all the modules as I am looking for complete information on evidence device or disk or system etc. and click next to proceed further.



In Add Data Source just click on Finish to generate the report of the device and you can perform complete investigate on the victim device or system or any other disk. It will process the data Source and add it to the local database.



- In Add Data Source just click on Finish to generate the report of the device and you can perform complete investigate on the victim device or system or any other disk.
- It will process the data Source and add it to the local database.
- After Process completion, it will show the Forensic Investigation Report.
- Now click on Devices Attached option, it will show the list of the attached device with the system.
- Now click on EXIF Metadata (Exchangeable image file format for images, sound used by Digital Camera, Smartphone and scanner), click on Installed Programs to see the entire installed programs in the system, Click Operating System Information.
- It will show the entire operating system list, Now Select Operating System User Account Option.
- It will Display the name of all the user Accounts,
- Now click on Recent Documents Option, it will display the latest created or opened documents,
- Click Web Bookmarks Option to see all the bookmarks by system users in different
- browsers, To see web cookies, select web cookies option, To See Web Downloads,
- Click on Web Downloads option, To check internet History, click on Web History Option, To see the history of internet search, click on Web Search Option, To see the list of all email ids in the system, click on email address.

Result:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

EXPERIMENT-03

Aim :

How to Recover Deleted Files using Forensics Tools

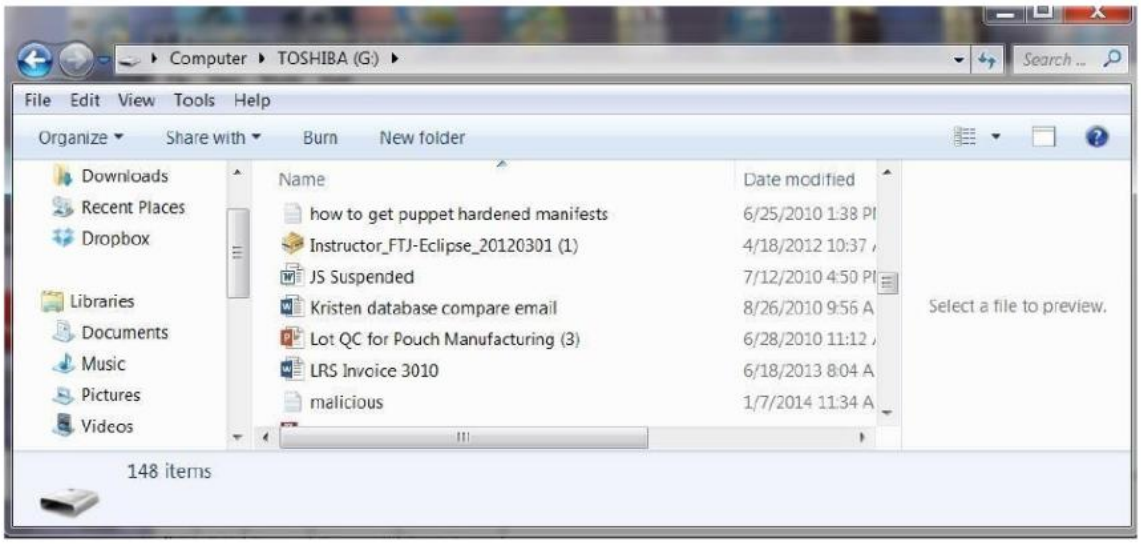
Step-01: Create a File

To demonstrate how to recover deleted files, let's create a malicious document. We will call this document "Malicious" and create it with Notepad in Windows



Step 2: Delete the File

Next, now that we have completed our plans to take over the world, let's delete the file because we no longer need it and we don't want to leave behind any evidence of our malicious plans.



Right-click on the malicious file and select delete.

Step 3: Create an Image

The first step a forensic investigator will do when examining your computer is to make a bit-by-bit copy of your hard drive or in this case your flash drive.

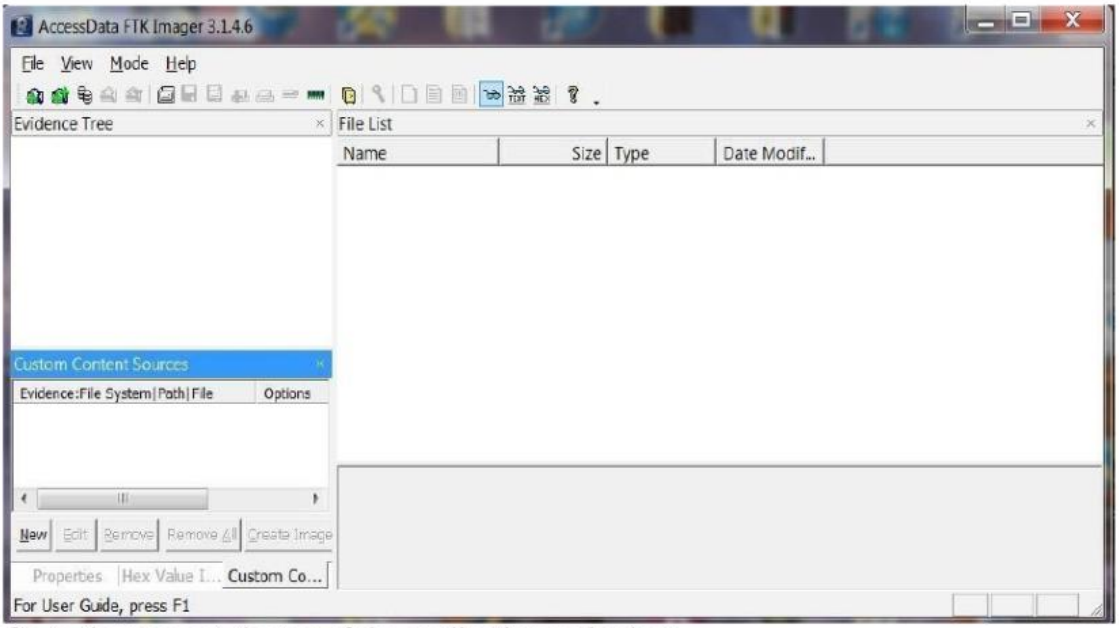
There are numerous tools that can do this and in Linux, we have the dd command that does an excellent job of making bit-by-bit copies (it's on all Linux distributions including BackTrack).

File backups and copies are not forensically sound as they will not copy deleted files and folders and in many cases will actually change the data.

Most forensic investigators use commercial tools. The two most popular being Encase by Guidance Software and Forensic Tool Kit by Access Data.

FTK, as it is commonly known in the industry, has a free imager that creates a bit-by-bit copy of the drive. This imager is probably the most widely used in the industry and its price is right, so let's use it

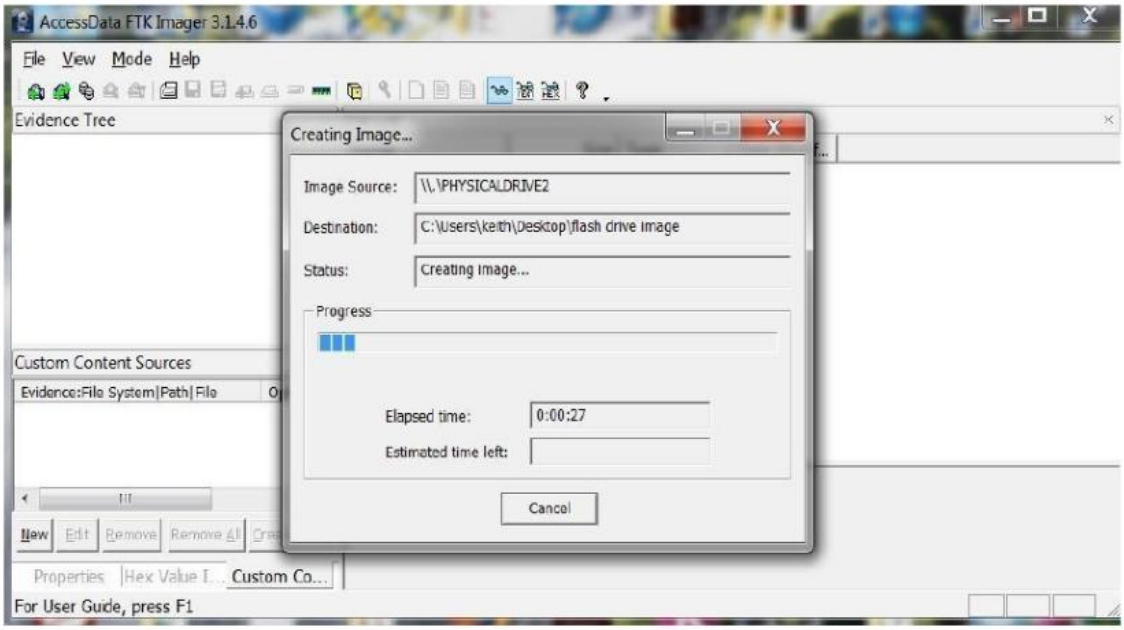
Now that have downloaded the **FTK imager**, we need to create a bit-by-bit image of the flash drive.



Go to the menu at the top of the application and select:

File -> Create Image

- It will open a wizard that will walk you through the process of opening a case and ask you for a case number, evidence number, examiner name, etc.
- Finally, it will ask for a location of the physical drive you want to image, a destination directory and a name for the image file.
- When you are done with all these administrative tasks, FTK Imager will begin the process of creating a forensically sound bit-by-bit image of your drive.



Now that we've created an image of the flash drive, we are ready to recover the deleted files.

Step 4: Recover Deleted Files

Once you have installed RecoverMyFiles, select the Start Recovery icon in the upper left corner.

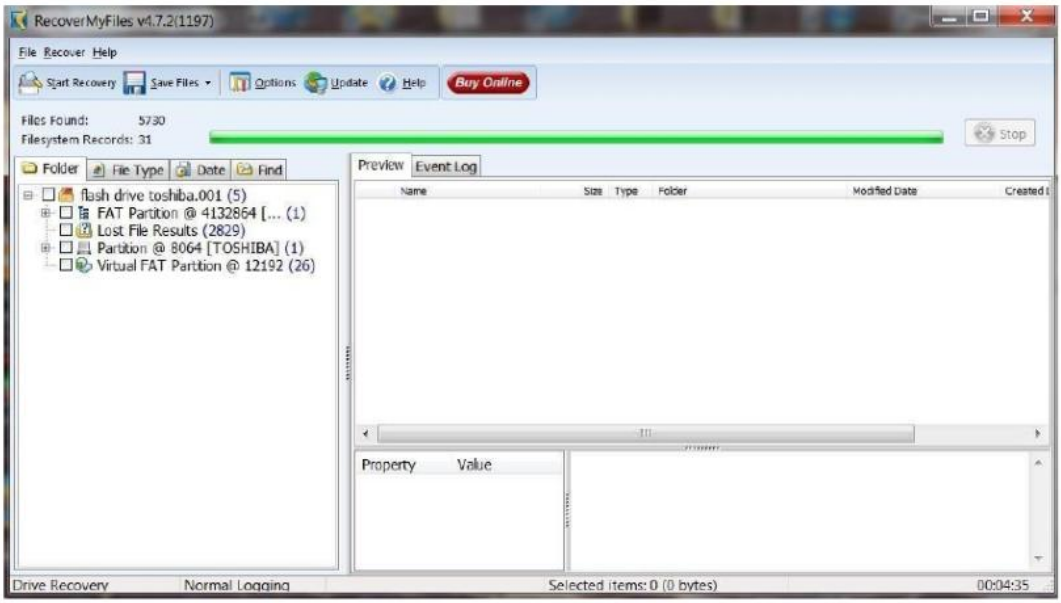
It will ask you to select either Recover Files or Recover Drive.

Select Recover a Drive. It will then search and display all your drives like that in the screenshot below.

Since we are using a forensic image, select Add Image button to the right. You will need to provide a path to your image file created with FTK

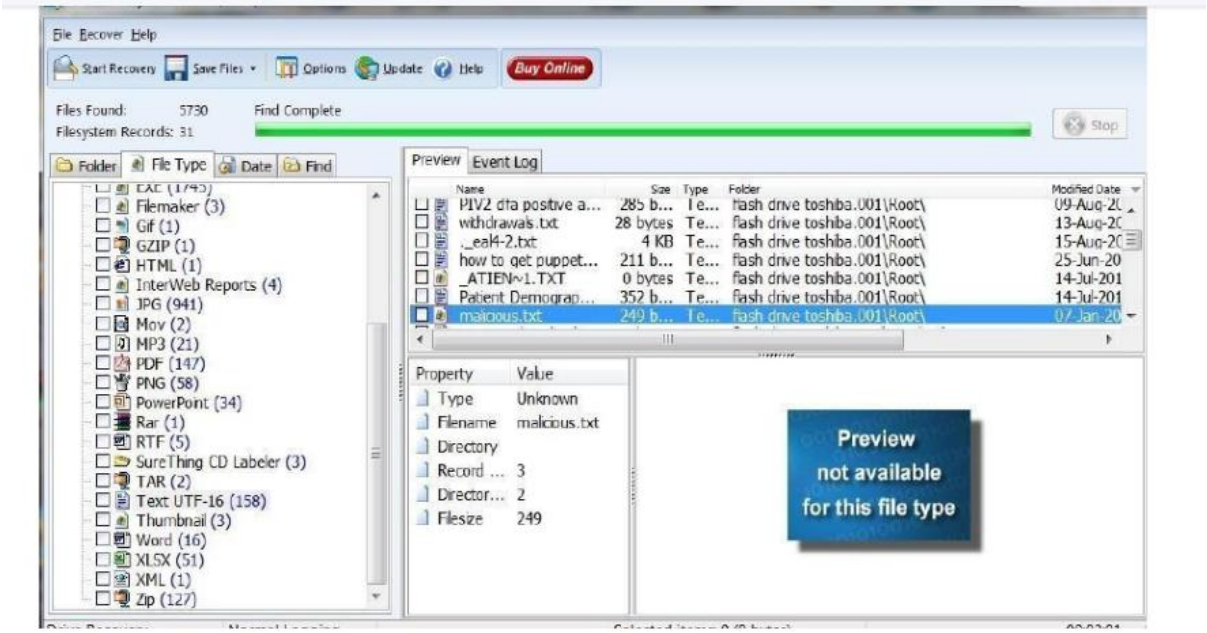


Once you select an image file, start the automatic file recovery. When the recovery is completed,



Once you select an image file, start the automatic file recovery. When the recovery is completed,

- Then selected the File Type tab above the Explorer window to categorize the files by type.
- There are numerous file types recovered from this flash drive.
- Since our malicious document was a .txt, I have selected the TXT UTF-16 file type.
- It then puts all 158 .txt files on display in the upper right window.
- It has recovered our malicious.txt file and everything on it. Busted!



Result:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

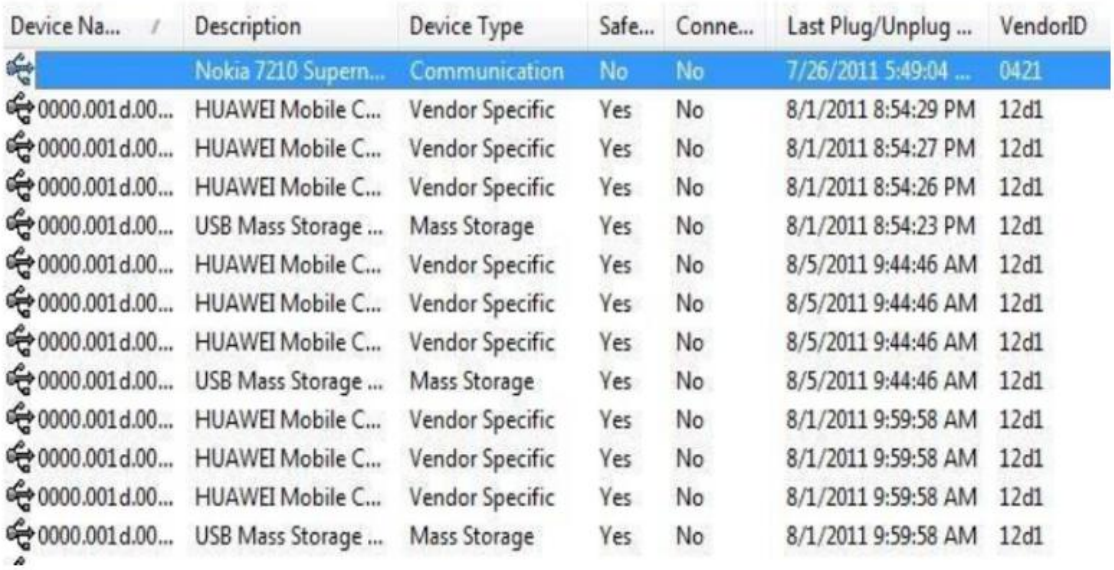
EXPERIMENT-04

Aim :

Find Last Connected USB on your system (USB Forensics)

Procedure :

- USBDeview is a small utility that lists all USB devices that currently connected to your computer, as well as all USB devices that you previously used.
- For each USB device, extended information is displayed: Device name/description, device type, serial number (for mass storage devices), the date/time that device was added, VendorID, ProductID, and more…
- USBDeview also allows you to uninstall USB devices that you previously used,disconnect USB devices that are currently connected to your computer, as well as to disable and enable USB devices.
- You can also use USBDeview on a remote computer, as long as you log in to that computer with admin user



Device Na...	Description	Device Type	Safe...	Conne...	Last Plug/Unplug ...	VendorID
	Nokia 7210 Supern...	Communication	No	No	7/26/2011 5:49:04 ...	0421
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/1/2011 8:54:29 PM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/1/2011 8:54:27 PM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/1/2011 8:54:26 PM	12d1
0000.001d.00...	USB Mass Storage ...	Mass Storage	Yes	No	8/1/2011 8:54:23 PM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/5/2011 9:44:46 AM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/5/2011 9:44:46 AM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/5/2011 9:44:46 AM	12d1
0000.001d.00...	USB Mass Storage ...	Mass Storage	Yes	No	8/5/2011 9:44:46 AM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/1/2011 9:59:58 AM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/1/2011 9:59:58 AM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/1/2011 9:59:58 AM	12d1
0000.001d.00...	USB Mass Storage ...	Mass Storage	Yes	No	8/1/2011 9:59:58 AM	12d1

Result:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

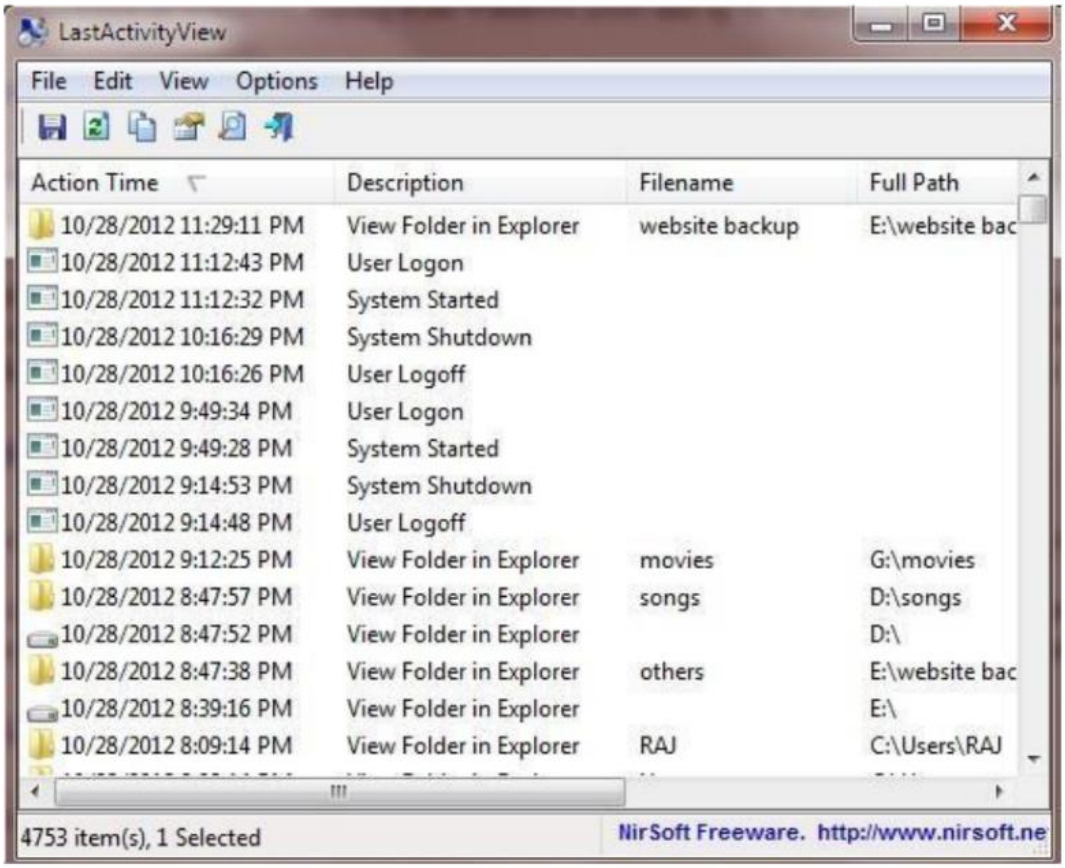
EXPERIMENT-05

Aim :

How to View Last Activity of Your PC

Procedure :

LastActivityView is a tool for Windows operating system that collects information from various sources on a running system, and displays a log of actions made by the user and events occurred on this computer.



Result:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

EXPERIMENT-06

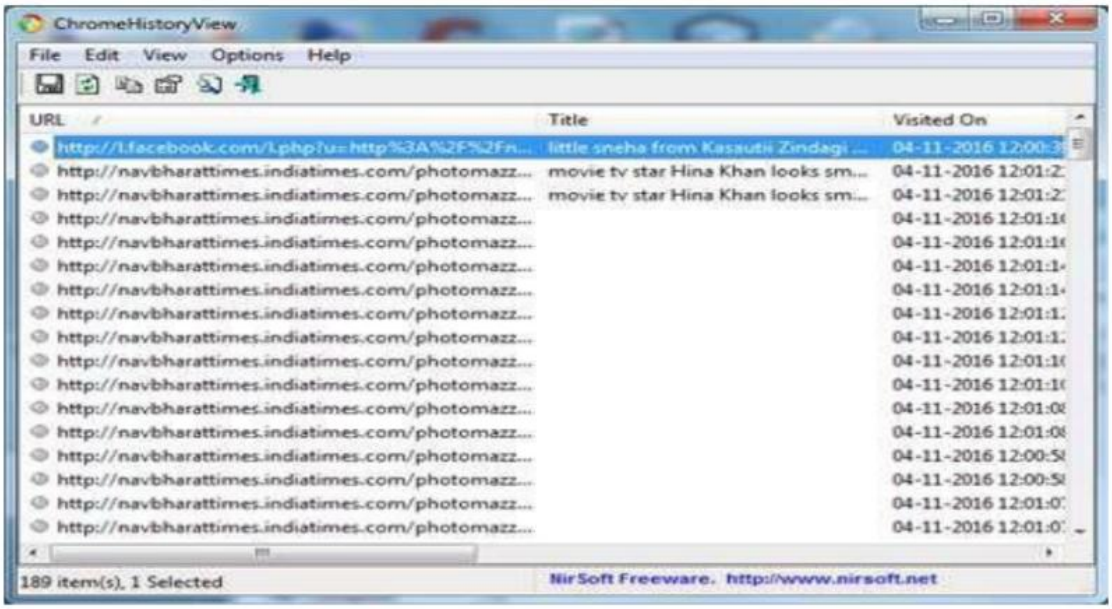
Aim :

How to Extracting Browser Artifacts

Procedure :

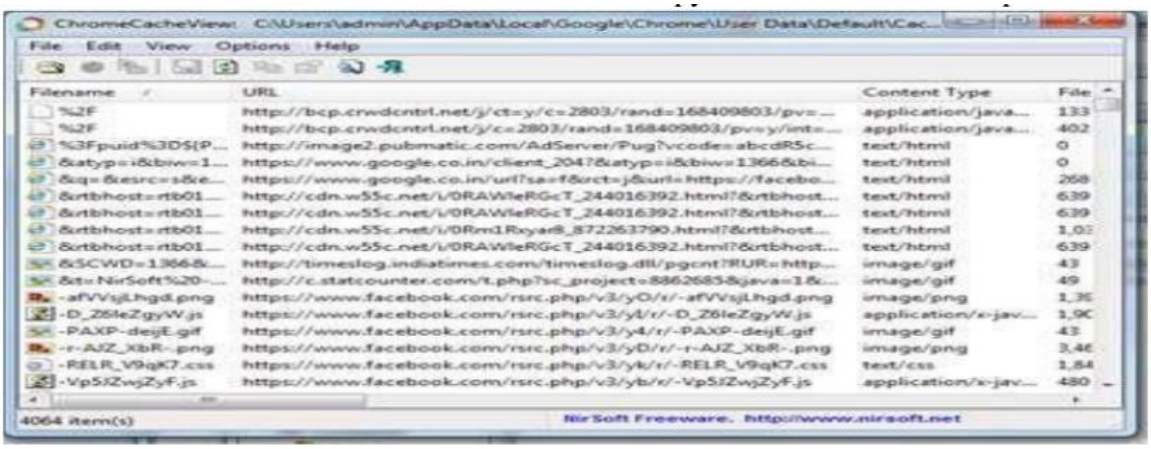
- ChromeHistoryView: is a small utility that reads the history data file of Google Chrome Web browser, and displays the list of all visited Web pages in the last days.

- For each visited Webpage, the following information is displayed: URL, Title, Visit Date/Time, Number of visits, number of times that the user typed this address (Typed Count), Referrer, and Visit ID



ChromeCacheView:

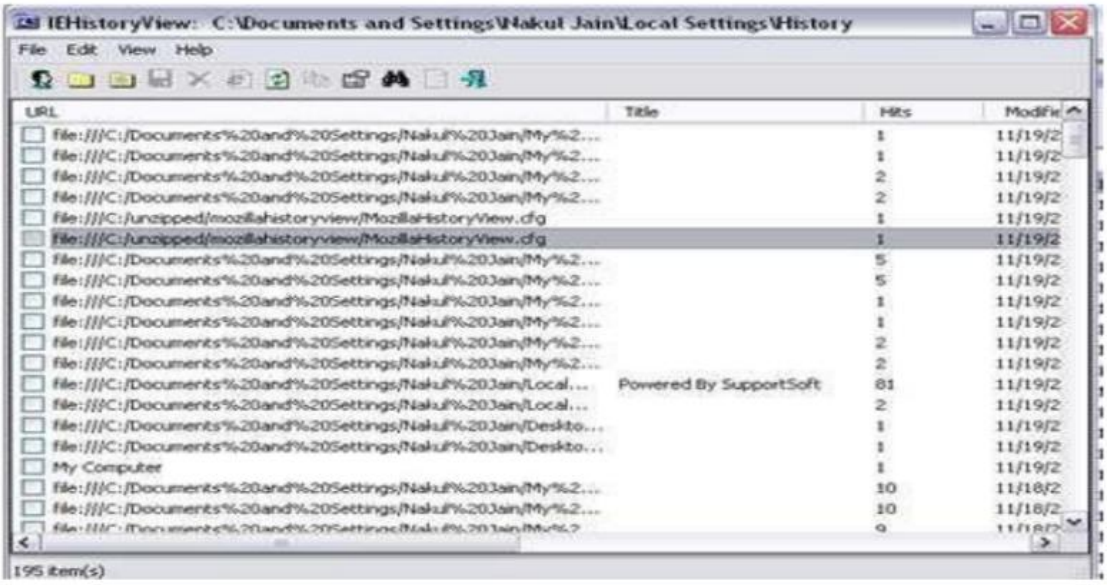
- ChromeCacheView is a small utility that reads the cache folder of Google Chrome Web browser, and displays the list of all files currently stored in the cache.
- For each cache file, the following information is displayed: URL, Content type, File size, Last accessed time, Expiration time, Server name, Server response, and more.



IEHistoryView:

- This utility reads all information from the history file on your computer, and displays the list of all URLs that you have visited in the last few days.

- It also allows you to select one or more URL addresses, and then remove them from the history file or save them into text, HTML or XML file.



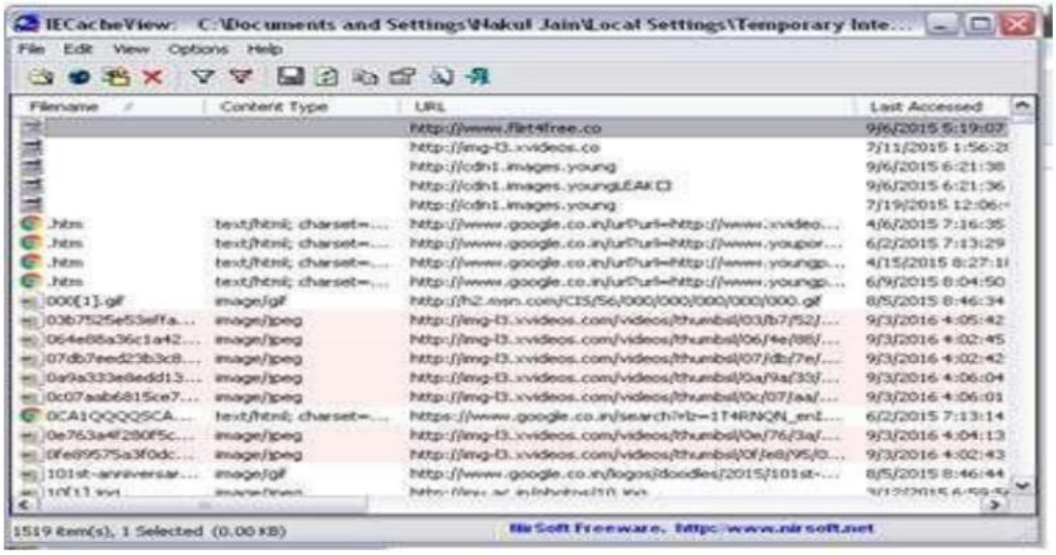
IEHistoryView:

- This utility reads all information from the history file on your computer, and displays the list of all URLs that you have visited in the last few days.
- It also allows you to select one or more URL addresses, and then remove them from the history file or save them into text, HTML or XML file.

IECacheView:

- IECacheView is a small utility that reads the cache folder of Internet Explorer, and displays the list of all files currently stored in the cache.

- For each cache file, the following information is displayed: Filename, Content Type, URL, Last Accessed Time, Last Modified Time, Expiration Time, Number of Hits, File Size, Folder Name, and full path of the cache filename.



Result:
Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

EXPERIMENT-07

Aim :

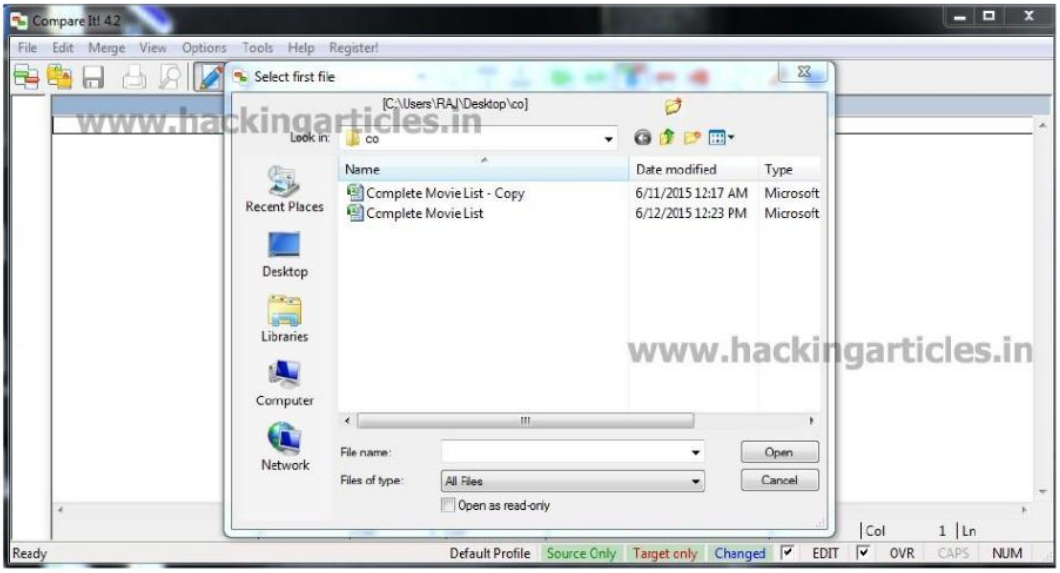
Comparison of two Files for forensics investigation by Compare IT software

Procedure :

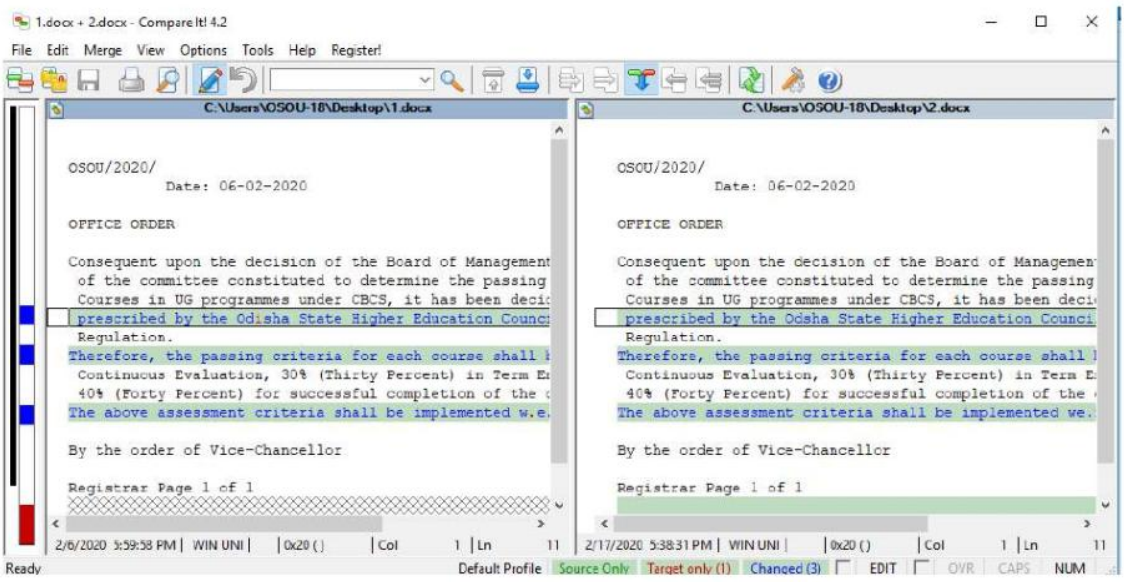
- Compare It is a software that displays 2 files side by side, with colored differences sections to simplify analyzing. You can move changes between files with a single mouse click or keystroke, and of course, you have the ability to edit files directly in comparison window. It can make colored printout of differences report, exactly as it's on the screen.
- Install the Compare It Software.

- Click on Compare It Tool, It will show a window to select the files to be compared.

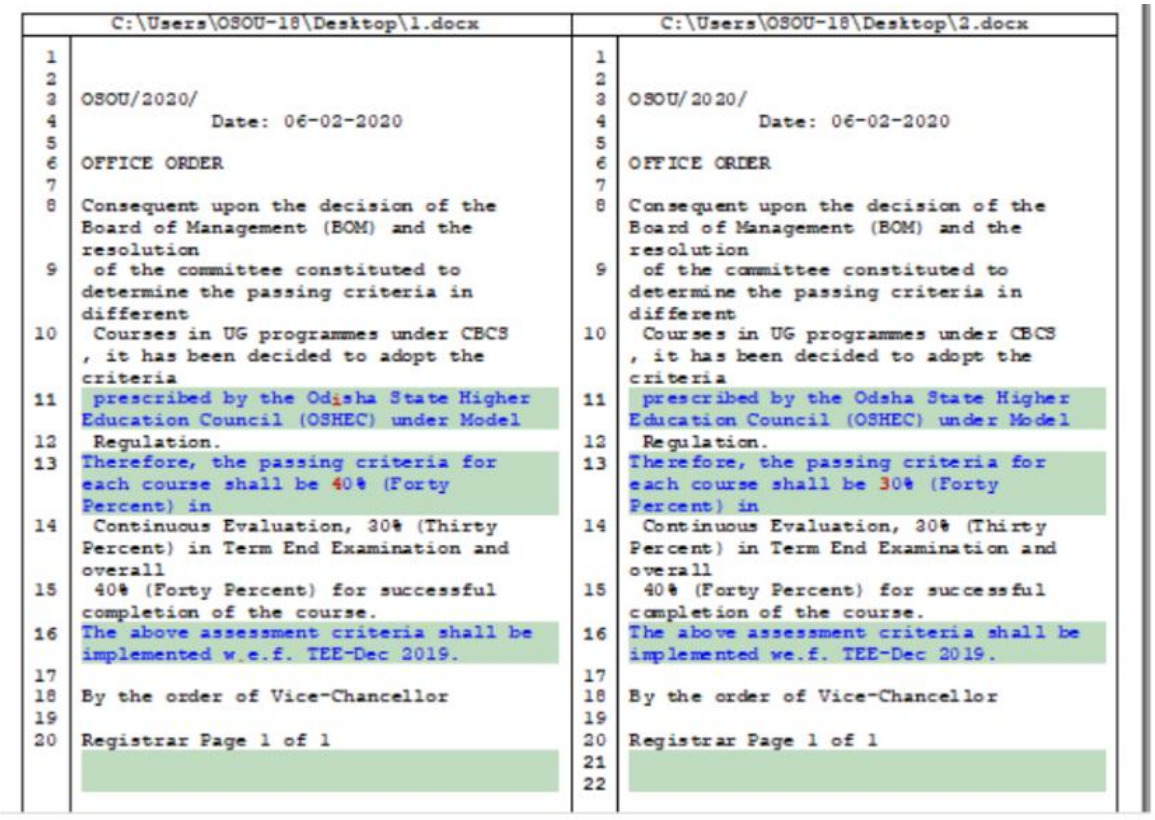
- First, select the first file and click on open and then select the second file and click on open.



Now it will show us the changes in the highlighted bar.



It also gives you Print report of the difference in the file as follows



Result:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

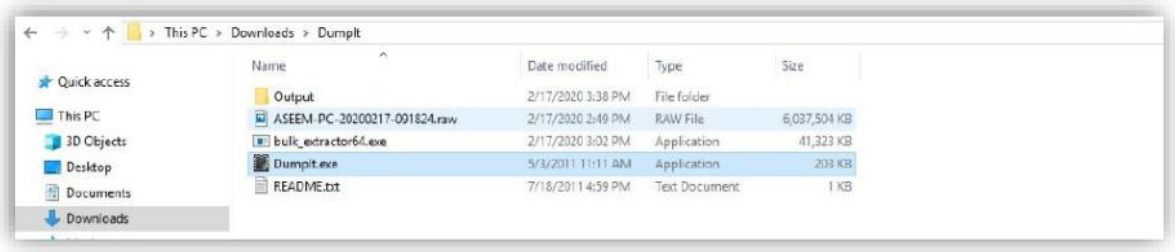
EXPERIMENT-08

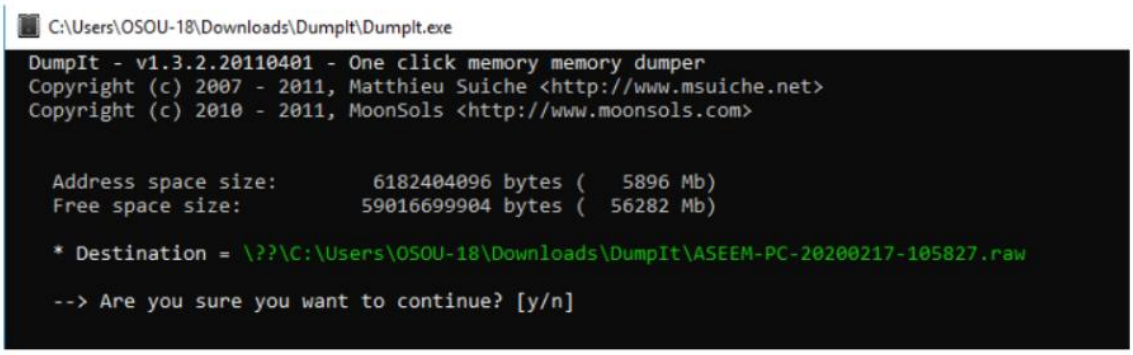
Aim :

How to Collect Email Evidence in Victim PC

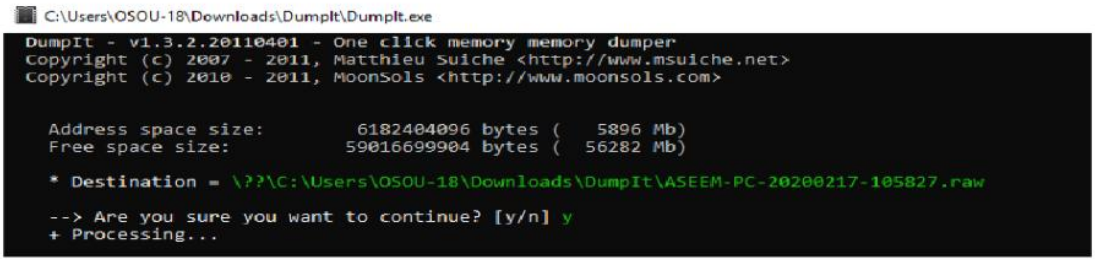
Procedure :

- To collect email evidence from Victim PC the first step is to capture the victim's RAM. This can be possible using **dumpit** tool.
- This utility is used to generate a physical memory dump of Windows machines.
- It works with both x86 (32-bits) and x64 (64-bits) machines. The raw memory dump is generated in the current directory, only a confirmation question is prompted before
- starting.
- Perfect to deploy the executable on USB keys, for quick incident responses needs.
- Run Dumpit.exe file the raw memory dump will be generated and save to the same directory

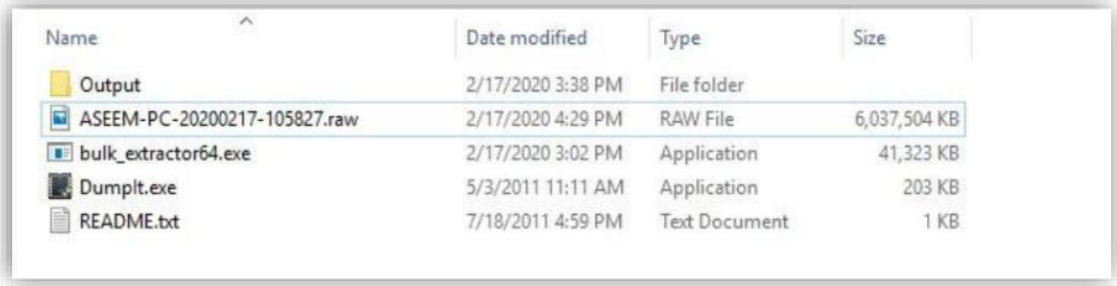




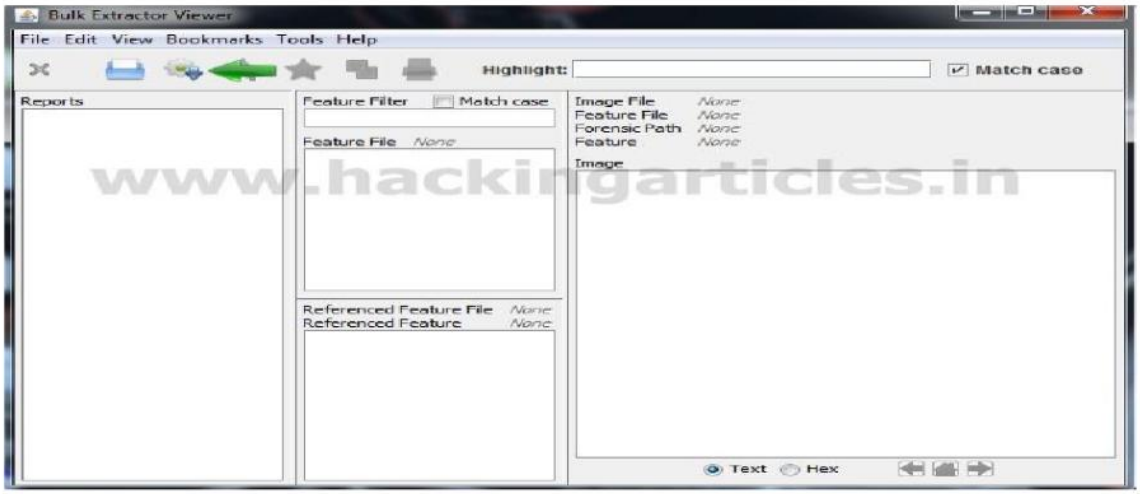
Write 'Y' for processing



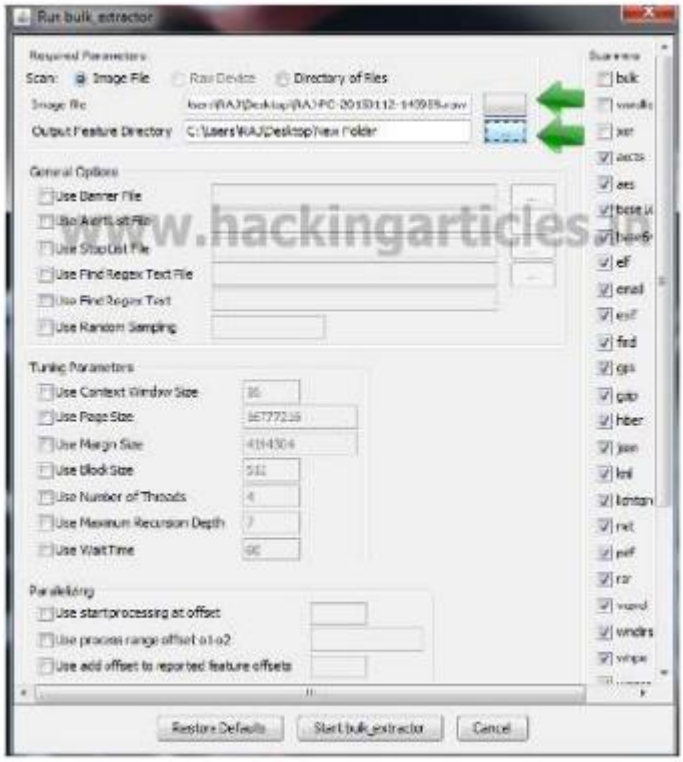
The output .RAW file will be as follows



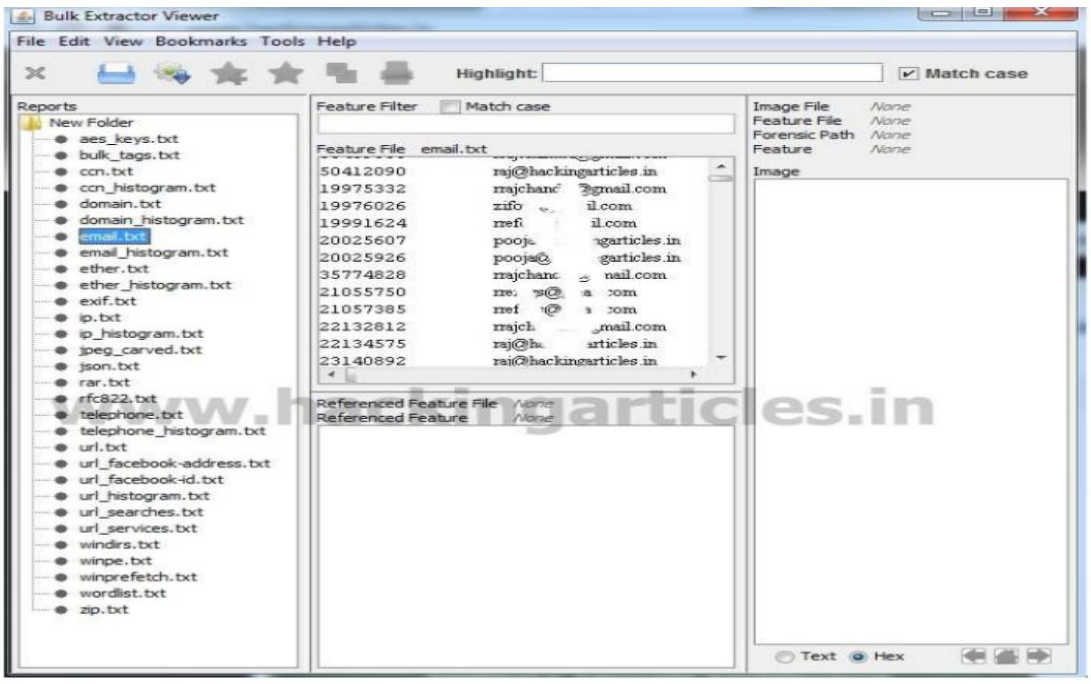
Then Download bulk extractor viewer from GitHub and install it in your PC. Now open bulk extractor viewer and click on to generate report.



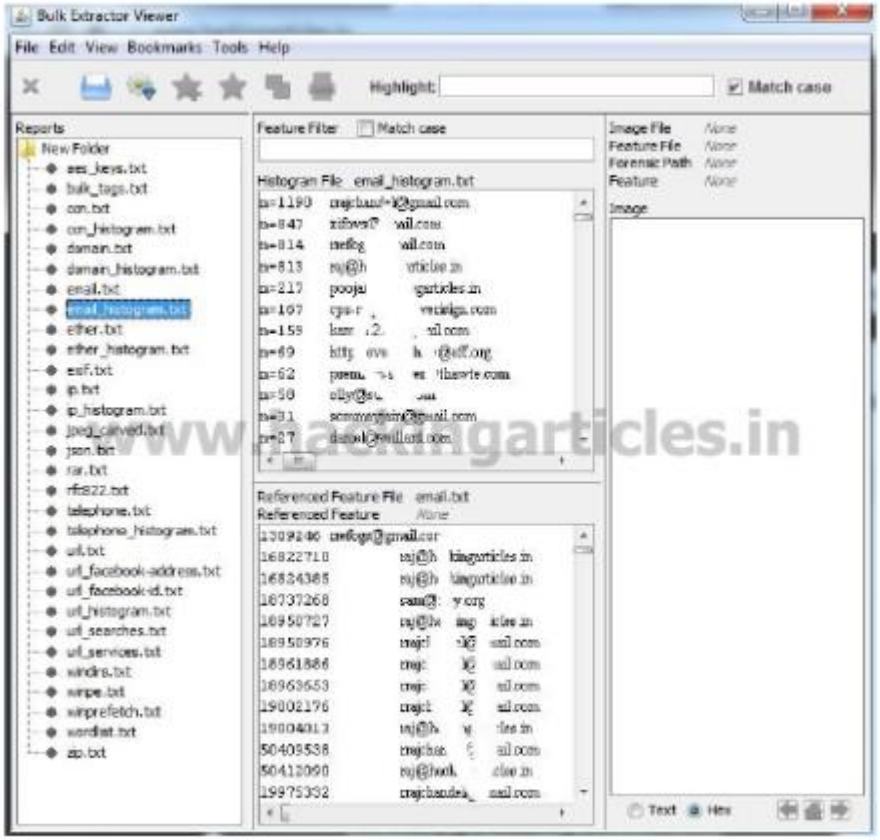
Now select the dump it image file and select an output folder for the report and click on start bulk extractor as seen below



Now in order to investigate the victim saved information of Email ID Click on email.txt as seen Below.



And also click on email_histogram.txt



Result:
Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

EXPERIMENT-09

Aim :

To study the steps for hiding and extract any text file behind an image file/ Audio file using Command Prompt.

Procedure :

Any file like .rar .jpg .txt or any file can be merged inside another file. In a simple way, we shall learn how to hide a text file inside an image file using the Command Prompt.

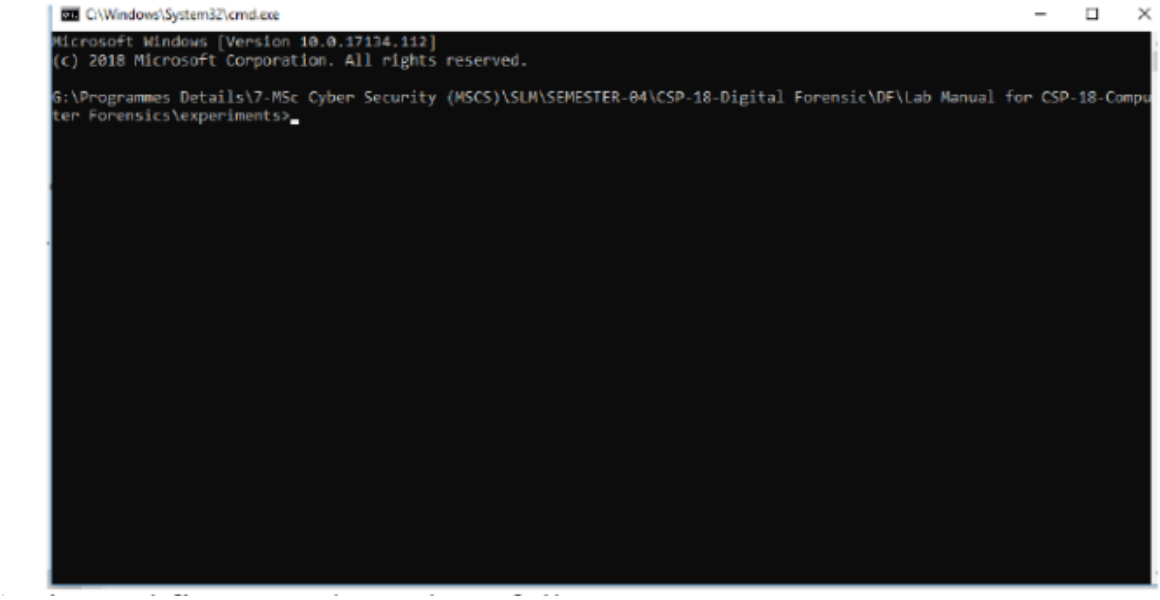
How to Hide the FILE?

Suppose you have to hide a text file “A.txt” with the image file “B.jpg” and combine them in a new file as “C.jpg”.Where “C.jpg” is our output file which contains the text hidden in the image file.



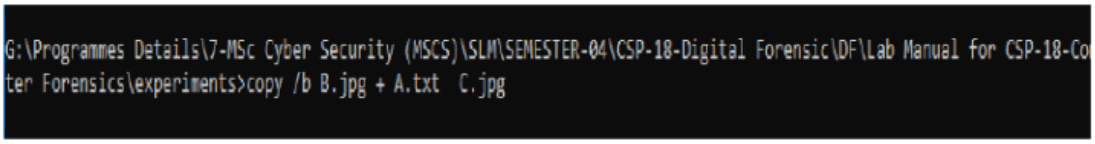
Follow the steps:

- Copy the file,u need to hide, to desktop(for our tutorial let us assume the file to be "A.txt")
- Copy the image, within which you need to hide the file, to desktop (let it be "B.jpg")
- Now open the cmd:
- **>ctrl+r**

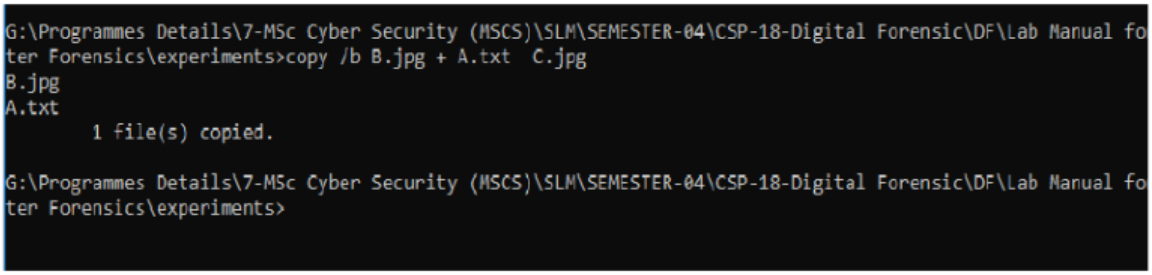


In cmd first type the code as follows:

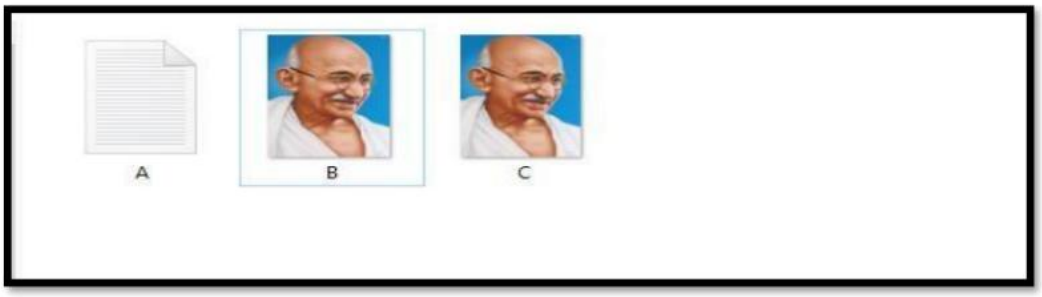
- **>cd desktop**
- NOTE: this code is for assigning the location on cmd to desktop
  Now type the following code:
  - **copy /b B.jpg + A.txt C.jpg**



Syntax: copy /b Name-of-file-containing-text-you-want-to-hide.txt + Name-of-initial-image.jpg Resulting-image-name.jpg

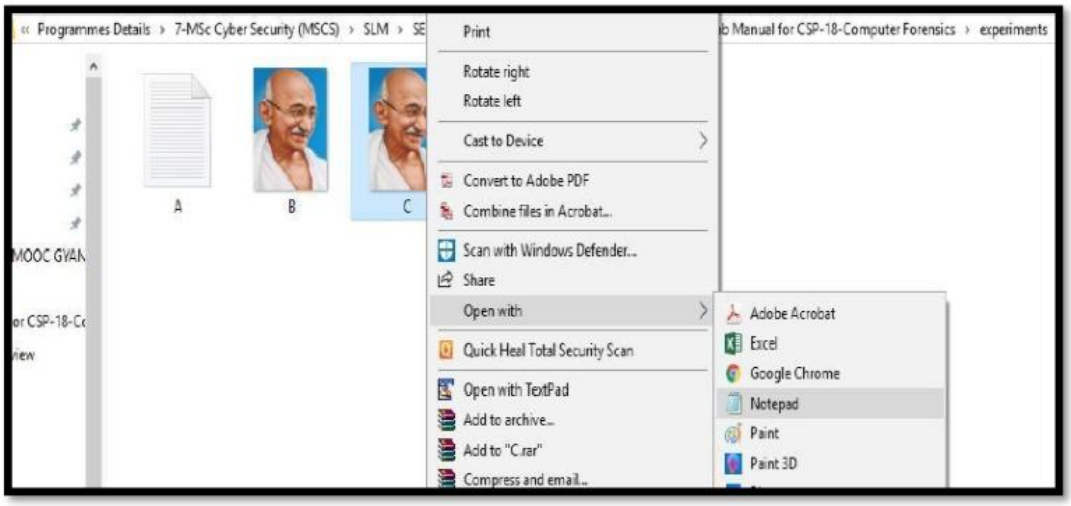


"C.jpg" is the output image inside this out image our file is hidden



How to retrieve the file?

1. locate C.jpg file from where you want to retrieve text data
2. Right-click and open with notepad



Done! Successfully opened! In the last of the notepad, you'll find the content of the text file



Hide A Message Into Image:

- Open Run command window by pressing win + r.
- Open command prompt by typing cmd and press OK
- Enter the directory where you have your files.
- Then type the command :echo "Your Message">>"image.jpg"
- Now the message is successfully hidden in the image file.
- To view the message: Open with Notepad, at last, you'll find the Your Message
- echo "Your Message">>"image.jpg"
- Now the message is successfully hidden in the image file.

To view the message:

Open with Notepad, at last, you'll find the Your Message

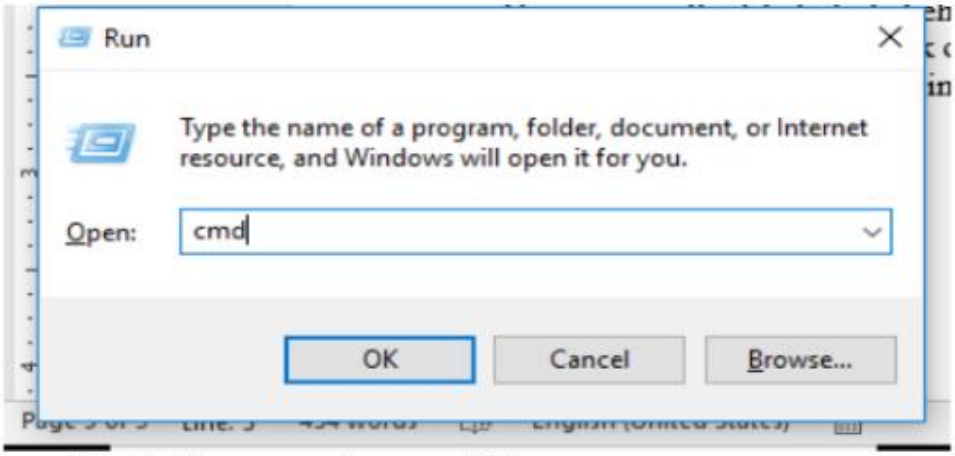
Another Method

1. Open Run command window by pressing win + r.

2. Open command prompt by typing cmd and press OK

3. Enter the directory where you have your files.

4. Then type the command



>> copy /b B.jpg + A.rar C.jpg

Here a.rar is the file to hide behind the image file (b.jpg) and the output file is c.jpg. To view the RAR file: right-click on the output image (here, c.jpg) and open with WinRAR. You'll find the file inside the image.

Hide File and text behind Audio File

Firstly get hold of a sound file you want to hide the data in (example sound.mp3), then gather all your files you want to hide and put them in a ZIP (example secret.zip).

Our chosen Sound and zip file:



Windows 7/10: Shift+right click in the folder containing the files will open the command prompt in that directory Windows: Open command prompt (start->run cmd), then use cd to get to the folder where the files are stored.

Linux:

We now need to merge these files together, but we want to use a binary merge to keep the two files intact. With Windows copy command this uses the /B switch.

Linux Code:

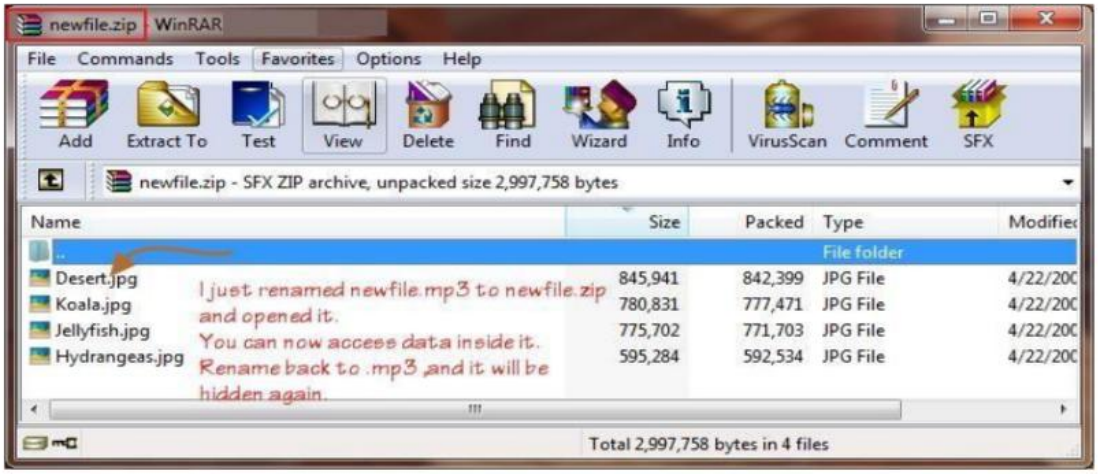
copy /b secret.zip + sound.mp3 newfile.mp3

cat sound.mp3 secret.zip > newfile.mp3

Here is the example sound containing a ZIP:



The two simplest ways to get your data back out of these files is to either change the extension from .mp3 to .zip or to open your chosen ZIP program and open newfile.mp3 within that. You should now be presented with your original files.



Result:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

EXPERIMENT-10

Aim:

How to Extract Exchangeable image file format (EXIF) Data
from Image Files using Exifreader Software.

Introduction:

In many cases when a computer, phone, or mobile device is seized for evidence, the system will have graphic images that might be used as evidence. Obviously, in some cases, these graphic images may be evidence such as in child pornography cases. Most digital devices "stamp" information on these graphic images that can tell us a lot about the who, what, when, and where the pictures were taken. This information is known as EXIF data and can very often be useful to the forensic investigator.

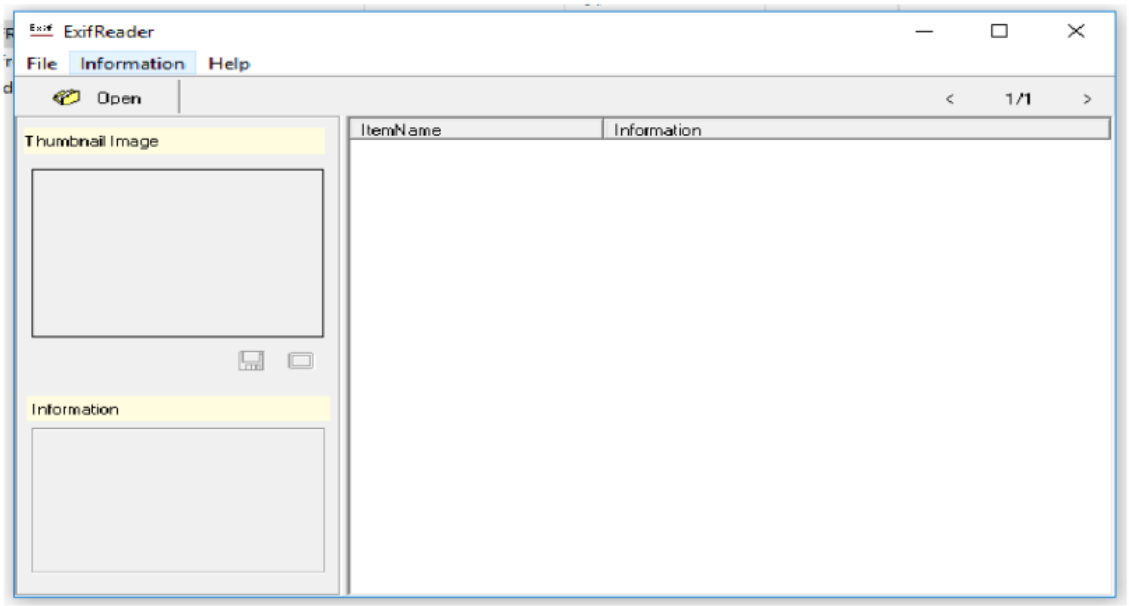
Exchangeable image file format (EXIF) is a standard set by the digital camera industry to identify formats for digital images and sound files. This information includes camera settings, time, date, shutter speed, exposure, whether a flash was used, compression, the name of the camera, and other information critical to viewing and editing the image in image-editing software. This information can be useful to the forensic investigator.

There are numerous applications that can extract this EXIF data from graphic files. Nearly every one of the major forensic suites (EnCase, FTK, Oxygen, etc.) has this capability built-in. For this lab, we will be using a simple, Windows-based tool called ExifReader (free).

Extract EXIF Data from Image Files

Step-01:

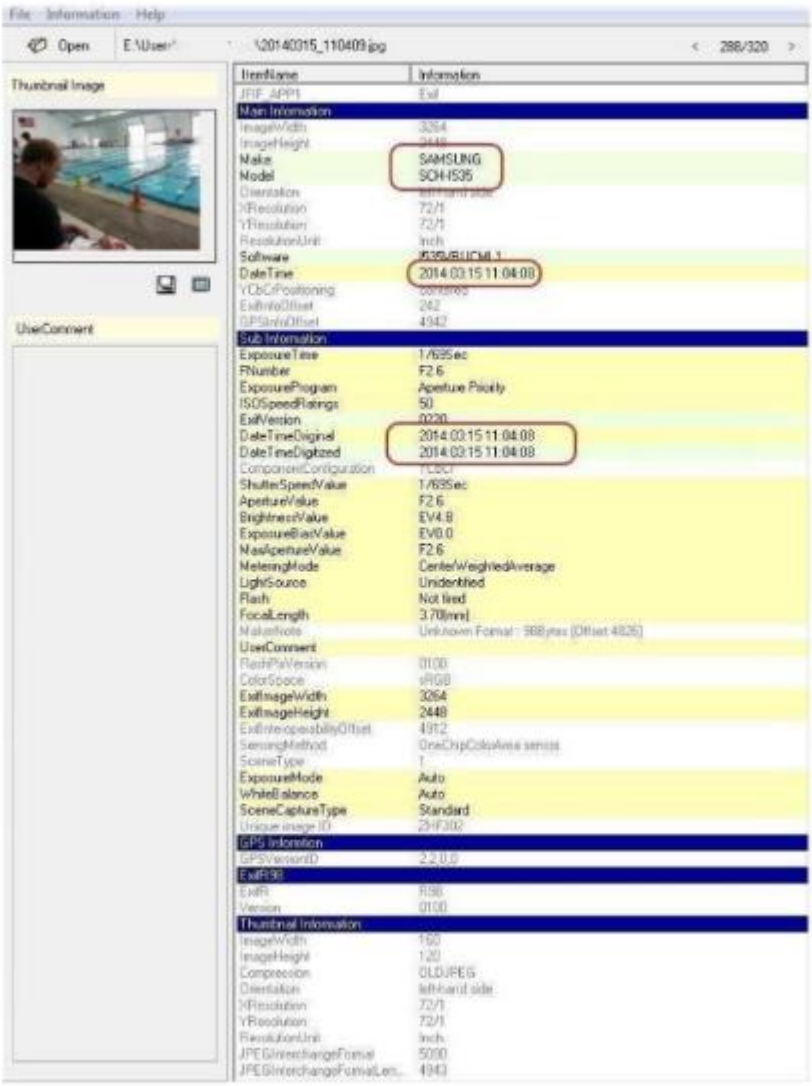
Download the ExifReader from the above link and click on the .exe file (ExifRead.exe) and it will open a clean and simple GUI Wizard as shown below:



Now, simply click on the "Open" button and browse to the pictures from the system or media. Normally, JPEG and JPG contain the maximum information, so let's use a JPEG file

Step-02: Open a Picture File

Once the selected picture opens the picture, it will load the picture into the thumbnail to the left and display the EXIF data to the right down the page as shown below.



There are lots of information you can collect in the EXIF data, but most are related to the technical specifications of the camera and photography. GPS coordinates of where the picture was taken. Most of this is of limited value to the forensic investigator.

Result:
Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.